

Memo Echo | 内部消息规范

文档摘要：本规范定义了系统内部微服务之间通过 MQ 进行异步通信的事件契约。各微服务必须严格按照约定的 Topic、Tag 和 Payload 结构进行生产与消费。

0. 全局 Payload 规范 (Base Envelope)

为保证文档排版紧凑与高可读性，本规范已提取所有 MQ 消息的公共外壳。下文各个独立事件中，**仅展示 data 字段内部的核心业务载荷 (Data Payload)**。

所有发往 MQ 的真实消息，均需包裹在以下标准外壳中：

```
{
  "eventId": "uuid-1234-5678",           // 事件唯一追踪ID，用于链路追踪
  "timestamp": 1712345678000,           // 事件发生时间戳
  "data": {
    /* 各事件具体业务字段，详见下文 */
  }
}
```

1. 群消息转日程

1.1 群聊消息待处理事件

- **事件描述:** bot_gateway 收到群聊消息，发送给 sensitive_filter 模块进行综合过滤。
- **发布方 (Producer):** bot_gateway
- **订阅方 (Consumer):** sensitive_filter
- **路由坐标:**
 - **Topic / Exchange:** topic_bot_messages
 - **Tag / RoutingKey:** group_msg_received
- **业务载荷 (Data Payload):**

```
{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "group",
  "sub_type": "normal",
  "message_id": -192837465,
  "group_id": 987654321,
  "user_id": 1122334455,
  "raw_message": "明天下午三点在三教开会",
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}
```

1.2 过滤后的群消息传递

- **事件描述:** `sensitive_filter` 过滤以及预处理消息后发送给 `ai_brain` 模块进行最后的日程录入。
- **发布方 (Producer):** `sensitive_filter`
- **订阅方 (Consumer):** `ai_brain`
- **路由坐标:**
 - **Topic / Exchange:** `topic_bot_messages`
 - **Tag / RoutingKey:** `group_msg_filtered`
- **业务载荷 (Data Payload):**

```
{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "group",
  "sub_type": "normal",
  "message_id": -192837465,
  "group_id": 987654321,
  "user_id": 1122334455,
  "raw_message": "明天下午三点在三教开会", //过滤后的消息
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}
```

1.3 大模型处理后的群日程

- **事件描述:** ai_brain 将处理后的日程发送给 persistence 模块进行日程登记。
- **发布方 (Producer):** ai_brain
- **订阅方 (Consumer):** persistence
- **路由坐标:**
 - **Topic / Exchange:** topic_bot_messages
 - **Tag / RoutingKey:** group_msg_extracted
- **业务载荷 (Data Payload):**

```
{
  "original_msg_id": -192837465, // 溯源ID
  "schedule_time": [1712401200000, 1712401200000], // 提取出的具体时间戳
  "location": "三教", // 提取出的地点
  "event_content": "开会", // 提取出的事件
  "participants": "全体同学", // 参与者
  "introduce": "为了庆祝朱宇翔生日, 学校邀请..."
}
```

1.4 返回给数据库的敏感词预警

- **事件描述:** sensistive_filter 将处理后的敏感信息发送给 persistence 模块进行敏感信息存储。

- 发布方 (Producer): `sensitive_filter`
- 订阅方 (Consumer): `persistence`
- 路由坐标:
 - Topic / Exchange: `topic_bot_messages`
 - Tag / RoutingKey: `group_msg_unsafe`
- 业务载荷 (Data Payload):

```
{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "group",
  "sub_type": "normal",
  "message_id": -192837465,
  "group_id": 987654321,
  "user_id": 1122334455,
  "raw_message": "明天下午三点在三教开会", // 大模型处
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}
```

2. 群用户提取日程

2.1 群聊@bot日程处理

- 事件描述: `bot_gateway` 收到群聊消息，发送给 `sensitive_filter` 模块进行需求提取。
- 发布方 (Producer): `bot_gateway`
- 订阅方 (Consumer): `sensitive_filter`
- 路由坐标:
 - Topic / Exchange: `topic_group_needs`
 - Tag / RoutingKey: `group_msg_received`
- 业务载荷 (Data Payload):

```
{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "group",
  "sub_type": "normal",
  "message_id": -192837465,
  "group_id": 987654321,
  "user_id": 1122334455,
  "raw_message": "[CQ:at,qq=123] 明天几点开会",
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}
```

2.2 过滤后的群日程提取需求

- **事件描述:** `sensitive_filter` 过滤以及预处理消息后发送给 `ai_brain` 模块进行日程查询语句编写。
- **发布方 (Producer):** `sensitive_filter`
- **订阅方 (Consumer):** `ai_brain`
- **路由坐标:**
 - **Topic / Exchange:** `topic_group_needs`
 - **Tag / RoutingKey:** `group_msg_filtered`
- **业务载荷 (Data Payload):**

```

{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "group",
  "sub_type": "normal",
  "message_id": -192837465,
  "group_id": 987654321,
  "user_id": 1122334455,
  "raw_message": "明天下午三点在三教开会", //过滤后的消息
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}

```

2.3 大模型处理后的群日程查询语句

- **事件描述:** ai_brain 将处理后的日程查询语句送给 persistence 模块进行日程查询。
- **发布方 (Producer):** ai_brain
- **订阅方 (Consumer):** persistence
- **路由坐标:**
 - **Topic / Exchange:** topic_group_needs
 - **Tag / RoutingKey:** group_msg_processed
- **业务载荷 (Data Payload):**

```

{
  "type" : "sql", // or "fun"
  "query" :{
    "sql" : "select * from .... "
  }
}

```

2.4 返回给Bot群日程

- **事件描述:** persistence 将处理后的日程发送给 bot_gateway 模块进行发送。
- **发布方 (Producer):** persistence
- **订阅方 (Consumer):** bot_gateway
- **路由坐标:**

- **Topic / Exchange:** `topic_group_needs`
- **Tag / RoutingKey:** `group_msg_result`
- **业务载荷 (Data Payload):**

```
{
  "type" : "user",
  "group_id": "10111",
  "schedules" :[
    {
      "original_msg_id": -192837465, // 溯源ID
      "schedule_time": [1712401200000, 1712401200000],
      "location": "三教",
      "event_content": "开会",
      "participants": "全体同学",
      "introduce" : "为了庆祝朱宇翔生日，学校邀请..."
    }
  ]
}
```

3. 私聊用户提取日程

3.1 私聊bot日程处理

- **事件描述:** `bot_gateway` 收到私聊消息，发送给 `sensitive_filter` 模块进行需求提取。
- **发布方 (Producer):** `bot_gateway`
- **订阅方 (Consumer):** `sensitive_filter`
- **路由坐标:**
 - **Topic / Exchange:** `topic_user_needs`
 - **Tag / RoutingKey:** `private_msg_received`
- **业务载荷 (Data Payload):**

```
{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "private",
  "sub_type": "normal",
  "message_id": -192837465,
  "user_id": 1122334455,
  "raw_message": "明天几点开会",
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}
```

3.2 过滤后的私聊日程提取需求

- **事件描述:** `sensitive_filter` 过滤以及预处理消息后发送给 `ai_brain` 模块进行日程查询语句编写。
- **发布方 (Producer):** `sensitive_filter`
- **订阅方 (Consumer):** `ai_brain`
- **路由坐标:**
 - **Topic / Exchange:** `topic_user_needs`
 - **Tag / RoutingKey:** `private_msg_filtered`
- **业务载荷 (Data Payload):**


```

{
  "time": 1712345678,
  "self_id": 10001000,
  "post_type": "message",
  "flag": null,
  "message_type": "private",
  "sub_type": "normal",
  "message_id": -192837465,
  "user_id": 1122334455,
  "raw_message": "明天下午三点在三教开会", //过滤后的消息
  "sender": {
    "user_id": 1122334455,
    "role": "member"
  }
}

```

3.3 大模型处理后的私聊日程查询语句

- **事件描述:** ai_brain 将处理后的日程查询语句送给 persistence 模块进行日程查询。
- **发布方 (Producer):** ai_brain
- **订阅方 (Consumer):** persistence
- **路由坐标:**
 - **Topic / Exchange:** topic_user_needs
 - **Tag / RoutingKey:** private_msg_processed
- **业务载荷 (Data Payload):**

```

{
  "type" : "sql", // or "fun"
  "query" :{
    "sql" : "select * from .... "
  }
}

```

3.4 返回给Bot私聊日程

- **事件描述:** persistence 将处理后的日程发送给 bot_gateway 模块进行发送。
- **发布方 (Producer):** persistence
- **订阅方 (Consumer):** bot_gateway
- **路由坐标:**
 - **Topic / Exchange:** topic_user_needs

- **Tag / RoutingKey:** private_msg_result

- **业务载荷 (Data Payload):**

```
{
  "type" : "user",
  "user_id": "10111",
  "schedules" :[
    {
      "original_msg_id": -192837465, // 溯源ID
      "schedule_time": [1712401200000, 1712401200000],
      "location": "三教",
      "event_content": "开会",
      "participants": "全体同学",
      "introduce" : "为了庆祝朱宇翔生日，学校邀请..."
    }
  ]
}
```